

Nikolas Espinoza

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EDUCATION

University of California, San Diego

La Jolla, CA

B.S. in Computer Science

Aug 2025

- **Coursework:** Intro to Computational Problem Solving, Vector Calculus, Intro to MATLAB, Data Structures and Object-Oriented Design

Downey High School

Downey, CA

Secondary Education

Aug 2021 - June 2025

- **Coursework:** AP Physics 1, AP Calculus BC, AP Environmental Science, AP Language and Composition

EXPERIENCE

Summer Internship Preparatory Program

La Jolla, CA

ECE Summer Intern

July 2025 – Present

- Collaborated with 3 members in circuit-based projects implementing several **CircuitPython** libraries and **30+ sensors and actuators**
- Practiced developing simple 3D models on **OnShape** and 2D/3D PCB designs with their schematics on **KiCad**

WorkWonder Apprenticeship

Downey, CA

Generative AI Apprentice

June 2024 - Aug 2024

- Utilized **PlayLab** to train a ChatBot capable of tutoring students in SAT-based topics and presented to other apprentices
- Designed a college budget utilizing **Excel's formula system**

PROJECTS

Point-A-Sketch - SIPP Hackathon

July 2025

- Worked alongside 3 other team members on an **embedded art program** that detects hand signals and interprets analog and sensor values as user controls
- Utilized **CircuitPython**, a **breadboard** and an **Adafruit M0 Express Board** to collect data from a **KY-037 sound sensor** to change the color value of the outputted line via the user's voice frequency
- Trained a **MLP regression model** with **OpenCV** and **ML** for gesture recognition, using **real-time pixel-level inference** to output the user's drawing direction

Unity-Based Video Game - SkillsUSA Competition Submission

September 2024 - June 2025

- Led a team of 2 developers to develop a fully-fledged video game deliverable with **20+ minutes of gameplay** that reached the **National competition levels, earning 6th place nationally**
- Used **Unity6's multiplayer capabilities** to develop a lobby system capable of connecting two remote computers with visibly changing UI and data in real time

ACTIVITIES AND LEADERSHIP

SkillsUSA Lead – Video Game Development Competition

September 2021 – June 2025

- Led new 2-4 member teams annually to develop a video game utilizing Unity's Game Engine
- Organized routine weekly meetings and deadlines to keep other members and myself on pace

High School Club Tutor

September 2021 – March 2025

- Guided **hundreds** of freshman students through homework and test review, including topics such as **transformations, geometric graphing, and algebra-based equations**

TECHNICAL SKILLS

Languages: C#, Java, Python, MATLAB

Developer Tools: Git, Unity, MATLAB, MaterialUI, MERN Stack

Libraries: Unity NetCode, NumPy, PySerial